

paper

by Aditya B

General metrics

32,648	4,335	258	17 min 20 sec	33 min 20 sec
characters	words	sentences	reading time	speaking time

Score

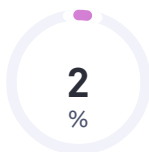


This text scores better than 84% of all texts checked by Grammarly

Writing Issues

101	10	91
Issues left	Critical	Advanced

Plagiarism



20 sources

2% of your text matches 20 sources on the web or in archives of academic publications

Writing Issues

86	Clarity	
42	Paragraph can be improved	
8	Intricate text	
36	Passive voice misuse	
12	Correctness	
2	Text inconsistencies	
6	Mixed dialects of english	
1	Misspelled words	
1	Incorrect punctuation	
1	Determiner use (a/an/the/this, etc.)	
1	Comma misuse within clauses	
3	Delivery	
2	Inappropriate colloquialisms	
1	Tone suggestions	

Unique Words

24%

Measures vocabulary diversity by calculating the percentage of words used only once in your document

unique words

Rare Words

44%

Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

rare words

Word Length

5.6

Measures average word length

characters per word

Sentence Length

16.8

Measures average sentence length

words per sentence

paper

```
\documentclass[conference]{IEEEtran} % or article

\usepackage{graphicx}
\usepackage{amsmath}
\usepackage{cite}

%opening
\title{Comparative Analysis of Electrical Activation Methods for Shape Memory Materials}
\author{Aditya Baskaran}

\begin{document}

\maketitle
\begin{abstract}
Shape memory alloys (SMAs) are materials that possess the ability to return to their original shape when exposed to stimuli such as temperature changes. This occurs due to a solid-to-solid phase transition. Properties such as Shape Memory Effect (SME) and superelasticity (SE) are also a result of this transition. In practice, the behavior of SMAs is non-linear and hysteretic, and their control is complex. Additionally, the scope of SMA application is limited by the slow recovery time after heating.
This paper compares different electrical activation methods for SMAs, including direct Joule heating, PWM, capacitor-discharge-based pulsed
```

heating, and electromagnetic methods such as induction. Each method is evaluated under similar conditions so that the comparison is fair, and the results are considered in terms of efficiency, response time, and how easy the system is to control.

The results show that PWM has the highest control, capacitor discharge has the fastest actuation, and induction offers the highest efficiency in regard to large-scale systems.

\end{abstract}

\section{Introduction}

Shape memory alloys (SMAs) are materials that can change shape and return to their original form when exposed to certain stimuli, most commonly temperature \cite{otsuka1999shape}. This behaviour occurs due to a solid-to-solid phase transformation between two different crystal structures, known as austenite and martensite \cite{lagoudas2008shape, brinson1993one}. Because of this, SMAs exhibit properties such as the shape memory effect (SME) and superelasticity (SE), which make them useful in applications like aerospace, robotics, and biomedical devices \cite{hartl2011review, kim2023shape}.

The development of SMAs dates back to the early 20th century, where Ölander first observed phase transformation behaviour in 1932 \cite{olander}. This was followed by Greninger in 1938, who observed the memory effect, which was later described more clearly by Vernon in 1941 as shape memory. However, SMAs only gained widespread attention in 1962 when Buehler et al. discovered the shape memory effect in Nickel-Titanium (NiTi), also known as nitinol \cite{pelton2000nitinol}. Since then, NiTi has become the most commonly used

SMA due to its availability, stability, and relatively good mechanical and thermomechanical properties.

However, in practice, the behaviour of SMAs is not straightforward. The response is non-linear and hysteretic \cite{shaw2008thermomechanical, liu1998hysteresis}, which makes it difficult to control accurately. In addition to this, the actuation speed is limited by the rate of heating and cooling, meaning that SMAs often have a relatively slow response compared to other actuator systems \cite{huang2012optimisation}. These limitations reduce their effectiveness in applications where fast or precise control is required.

To improve this, different electrical activation methods are used to control how heat is delivered to the SMA \cite{karunakaran2022heating}. Common methods include direct Joule heating, pulse-width modulation (PWM), capacitor-discharge-based pulsed heating, and electromagnetic approaches such as induction. Each of these methods affects the behaviour of the SMA in different ways, particularly in terms of efficiency, response time, and how easy it is to control.

This paper compares these activation methods under similar conditions in order to better understand their differences and identify which methods are more suitable for different types of applications. \cite{melton1980engineering, bellouard2008review, humbeeck1999sma}

\section{Mechanisms and Properties of SMA}

Shape memory alloys (SMAs) are unique active materials characterized by their ability to recover a predefined shape when subjected to specific thermal or

mechanical procedures. This behavior, known as the shape memory effect (SME), is driven by a reversible, diffusionless solid-to-solid phase transformation between two distinct crystal structures \cite{tanaka1986sma, lexcellent2004sma, lagoudas2008shape, otsuka1999shape}.

\subsection{Austenite vs. Martensite}

The mechanism of an SMA is defined by the interaction between its two primary phases \cite{brinson1993one}:²⁴

\begin{itemize}

\item \textbf{Austenite:} The parent phase, which is stable at high temperatures. It typically possesses a higher-symmetry crystal structure, such as the cubic B2 phase found in common NiTi alloys.

\item \textbf{Martensite:} The product phase, which is stable at lower temperatures. It has a lower-symmetry structure, which can be tetragonal, orthorhombic, or monoclinic (such as the B19' phase).

\item \textbf{Variants:} Martensite exists in two forms: twinned and detwinned. When an external force is applied to an SMA in its twinned martensitic state,²⁵ the variants reorient into detwinned martensite, resulting in a quasi-plastic deformation that remains after the load is removed.²⁶ Upon heating, this detwinned structure transforms back into the parent austenite phase, regaining the original shape.

\end{itemize}

\subsection{Transformation Temperatures}

The transformation between these two phases is governed by four characteristic temperatures:²⁷

`\begin{itemize}`

`\item \textbf{Martensite Start ($M_s$):}` The temperature at which the austenite begins to transform into twinned martensite during cooling.

`\item \textbf{Martensite Finish ($M_f$):}` The temperature at which the transformation to a fully martensitic phase is completed.²⁸

`\item \textbf{Austenite Start ($A_s$):}` The temperature at which the reverse transformation from martensite back to austenite begins upon heating.

`\item \textbf{Austenite Finish ($A_f$):}` The temperature at which the material returns completely to its parent austenitic state.

`\end{itemize}`

`\subsection{Transformation Temperatures}`

These temperatures are not fixed; they increase with applied mechanical stress²⁹ following the Clausius-Clapeyron relation `\cite{shaw2008thermomechanical}`.. For example, higher preloads require higher temperatures to initiate the SME because internal stress stabilizes the martensitic phase.

`\subsection{Hysteresis Loop}`

The transition between austenite and martensite does not occur at the same temperature during heating and cooling, resulting in a thermal hysteresis loop \cite{liu1998hysteresis, sun2000thermodynamics}.

\begin{itemize}

\item \textbf{Definition:} Hysteresis is the temperature difference between the heating and cooling paths, often characterized by the difference between the temperatures at which 50\% of the material has transformed during each cycle.

\item \textbf{Performance:} A narrow hysteresis is generally preferred for applications requiring fast reversibility and higher efficiency. Large hysteresis can lead to significant delays in response.

\item \textbf{Training:} To ensure reliable performance and a stable hysteresis loop, SMA actuators must be "trained" through repeated thermo-mechanical cycling under load until their behavior becomes predictable.

\end{itemize}

\cite{liu2004sma}

\subsection{Latent Heat}

Because the transformation is a first-order phase change, it involves the absorption or release of latent heat.

\begin{itemize}

\item \textbf{Thermal Dynamics:} During the forward transformation (cooling from austenite to martensite), the process is exothermic, meaning heat is

³⁰ released to the surroundings. Conversely, the reverse transformation (heating from martensite to austenite) is endothermic, requiring the material to absorb energy from its environment or an internal heat source.

\item \textbf{Impact on Control:} This energy exchange causes a thermal response lag, where a delay exists between the application of a heating current and the resulting force increase. In high-speed actuation, such as capacitor discharge, delivering energy rapidly enough to overcome this latent heat ³¹ threshold is essential for achieving adiabatic activation, which minimizes total ³¹ energy loss.\cite{miyazaki1989sma}

\end{itemize}

\section{Activation Methods}

\subsection{Direct Joule Heating}

Resistive heating or Direct Joule Heating is a simple and convenient mechanism to control SMA Activation.\cite{huang2012optimisation, karunakaran2022heating, zhang2014thermal} The electric power P is a function of the resistance R of the wire and the Current I flowing through it \eqref{eq:pow}. The current and voltage relation is given by Ohm's law ³² \eqref{ohmslaw}, and the resistance for a uniform material with resistivity ρ and constant current flow is given by \eqref{eq:rho}.

\begin{equation}

$$P = I^2 R$$

\label{eq:pow}

\end{equation}

\begin{equation}

$$I = \frac{V}{R}$$

\label{ohmslaw}

\end{equation}

\begin{equation}

$$R = \rho_{\text{R}} \frac{L}{A}$$

\label{eq:rho}

\end{equation}

The temperature of the wire can be predicted by its specific heat c ,
 $\text{\eqref{spheat}}$, where Q is the thermal energy, and in an ideal, adiabatic
 process where $\Delta Q = P \Delta t$, the temperature of the wire can be
 predicted by the equation $\text{\eqref{delT}}$. ³³ \cite{huang2012optimisation,
 karunakaran2022heating}

\begin{equation}

$$c = \frac{\Delta Q}{m \Delta T}$$

\label{spheat}

\end{equation}

\begin{equation}

$$\Delta T =$$

$$\left(\frac{16}{\pi^2} \right)$$

$$\left(\frac{\rho_{\text{R}}}{\rho_{\text{D}} d^4 c} \right)$$

$$(I^2 \Delta t)$$

\label{delT}

\end{equation}

It is worth noting here that this relationship assumes a constant current input. If instead the applied voltage is held constant, the current becomes inversely proportional to the wire length, $I = \frac{V}{R} \propto \frac{1}{l}$, leading to different thermal behavior.^{35 34} \cite{song2003heating, fan2005heat, zhou2010sma}

\subsection{PWM Control}

Pulse Width Modulation (PWM) is a control technique that utilizes a periodic train of electrical pulses where the base frequency remains fixed, but the width of each pulse is varied.^{36 36 36 36} In the context of shape memory alloys (SMAs), it is used to precisely regulate the amount of energy transferred to the material to control its temperature, length, and displacement.^{36 36}

Although it uses the same heat transfer principles as Direct Joule Heating, PWM controls the average power delivered through the duty cycle D ³⁹ \eqref{Ds}. Hence, the average power delivered to the SMA can be expressed as \eqref{pwmpow} where P_{on} would be the power delivered in the ON state.

\begin{equation}

$$D = \frac{t_{on}}{t_{on} + t_{off}}$$

\label{Ds}

\end{equation}

\begin{equation}

$$P_{\text{avg}} = D \cdot P_{on} = D \cdot I^2 R$$

\label{pwmpow}

\end{equation}

From \eqref{delT}, the temperature rise under Joule heating satisfies $\Delta T \propto I^2 \Delta t$. Incorporating PWM control, the temperature rise becomes

$$\Delta T \propto D I^2 \Delta t$$

demonstrating that the heating rate can be directly regulated⁴⁰ through the duty cycle. Furthermore, closed-loop control allows for feedback-based regulation, further improving precision.\cite{alalem2022pwm, ma2017control, ikuta1990micro, chung2008pwm, gupta2010control}

⁴¹
\subsection{Capacitor discharge/Pulsed Heating}Pulsed actuation using capacitor discharge achieves ultra-fast activation of SMAs by pulsing extremely high electrical currents through the SMA in a very short period of time (4.6 to 10ms)\cite{qiu2000high, tosunoglu2001actuation, lee2005fast}. Due to the high discharge, the SMA reaches its transformation temperature (A_f) before any substantial heat can dissipate, resulting in energy savings up to 80\% when compared to traditional quasi-static heating
\cite{motzki2018high}

The energy supplied by the capacitors E_t is the sum of the Sensible Heat (E_h) and the Latent Heat (E_p) of the SMA used. This can be given by⁴²

$$E_t = E_h + E_p = m \cdot (\lambda \Delta T + c)$$

The capacitor⁴¹ also has an exponential decay pattern, where the current at any time $I(t)$ ⁴³ can be expressed as

$$\begin{equation}$$

$$I(t) = \frac{V_0}{R} e^{-t/RC}$$

\end{equation}

where V_0 is the initial charged voltage, and C is the total capacitance. Due to the low resistance of SMAs \cite{qiu2000high}, the currents produced can exceed 3200A.

\cite{qiu2000high, motzki2018high, seelecke2010modeling, tosunoglu2001actuation, lee2005fast}

\subsection{Induction heating}

Induction heating for shape memory alloys (SMAs) is a contactless activation method that utilizes alternating current (AC) and electromagnetic fields to generate internal heat via eddy currents \cite{saunders2016validated}

This method is uniform in larger-scale applications and has the ability to be regulated precisely.

Induction utilizes Faraday's Law to induce eddy currents directly within the ⁴⁴ conductive SMA. This generates heat from within the material, eliminating the ⁴⁵ need for thermal penetration from an external medium.

Induction systems often feature feedback controllers with high temperature sensitivity, and because the heating is uniform and homogenous, induction-based SMAs produce stable and prominent hysteresis loops. ⁴⁶

Induction is also significantly more efficient than direct DC, particularly in large-scale applications. For example, generating a specific heating rate in a torque tube required only 50 A via induction, whereas direct DC heating would have required 370. \cite{saunders2016validated}

By varying the AC frequency, researchers can selectively activate different parts of a multi-functional composite. For instance, a 13.56 MHz field can trigger one section while a 296 kHz field triggers another. ⁴⁷

\cite{doyle2004thermal, lagoudas2012cooling}

\subsection{Magnetic Field}

Magnetic activation of shape memory materials is a contactless stimulus method primarily utilized for shape memory polymers (SMPs) and multi-functional composites. This method uses alternating magnetic fields to trigger shape recovery by generating heat directly within the material.

Magnetic activation relies on inductive heating, where the material's internal temperature is raised⁴⁸through three primary electromagnetic mechanisms:

Eddy Current Losses: Dominant in larger, electrically conductive components.

Hysteresis Losses: Common in micro-sized magnetic particles at relatively low frequencies.

Rotational Losses: The primary mechanism for magnetic nanoparticles under high-frequency alternating fields.

Because heat is generated⁴⁹ inside the polymer itself, magnetic activation allows⁵⁰ for a rapid temperature increase⁵⁰ with a reduced thermal gradient compared to external heating.

The main limitation with Magnetic SMA Activation is the material constraints. Since pure polymers are typically non-magnetic and non-conductive, magnetic responsiveness is achieved by embedding magnetic nanoparticles into the polymer matrix.⁵¹ \cite{mohr2006initiation, liu2017stimulus, karunakaran2022heating}

\section{Methodology}

In order to compare the different methods previously outlined in a fair manner, certain assumptions are made, which will be specified. Additionally, the SMA sample is assumed to be a NiTi wire in all cases except Magnetic-field-activation, where Ni-Mn-Ga is the most common alloy. \cite{doyle2004thermal, lagoudas2012cooling}

\subsection{Assumptions and Conditions}

\begin{itemize}

\item All simulations are conducted at a constant ambient temperature T_{env} .

\item The SMA is assumed to have uniform geometry and material properties throughout.

\item Heat transfer is modeled using a lumped thermal system \cite{huang2012optimisation, doyle2004thermal}, neglecting spatial temperature gradients.

\item Convective heat loss is assumed to be linear and proportional to the temperature difference.

\item Radiation losses and environmental disturbances, such as airflow and pressure variations, are neglected.

\item A single SMA material (NiTi) with consistent transformation temperatures is assumed across all methods (except Magnetic-Field-Activation).

\end{itemize}

\subsection{Compared Metrics}

The performance of each activation method is based on the following criteria:

\begin{itemize}

- \item \textbf{Efficiency:} Defined as the ratio of useful thermal energy contributing to phase transformation to the total electrical energy supplied.
 - \item \textbf{Response Time:} The time required for the SMA to reach the transformation temperature (A_s or A_f).
 - \item \textbf{Control and Precision:} The ability to precisely regulate temperature and, consequently, displacement, typically assessed through the smoothness and adjustability of the input signal.
- \end{itemize}

\subsection{Thermal Model}

The general equation for temperature change is given as ⁵⁶ \eqref{comp}, where $P(t)$ is the electrical power input for the SMA, h is the heat transfer coefficient, T is the temperature of the SMA, and T_{env} is the ambient temperature. Rearranging \eqref{comp} and taking final temperature as steady state (T_{ss}), we get \eqref{sstate}

\begin{equation}

$$\frac{dT}{dt} = \frac{P(t) - h(T - T_{env})}{m \cdot c}$$

\label{comp}

\end{equation}

\begin{equation}

$$T_{ss} = T_{env} + \frac{P}{h}$$

\label{sstate}

\end{equation}

$P(t)$ for some relevant conditions is given by:

```

\begin{itemize}
\item Direct:  $P = I^2 R$ 
\item PWM:  $P = D \cdot I^2 R$ 
\item Pulsed: Time-varying
\item Induction:  $P \propto B^2$ 
\end{itemize}

```

\section{Comparative Analysis}

Although SMA activation methods are primarily thermal in nature,⁵⁷ they vary in⁵⁷ how the heat is regulated and delivered.⁵⁷ This results in different outcomes when one SMA wire is activated using different methods.

\subsection{Efficiency:}

\textbf{Direct Joule Heating}: Traditional or "quasi-static" Joule heating is only moderately efficient because a large amount of heating energy is lost to the environment through heat exchange during the activation process. This low efficiency is a direct result of the long durations required to heat the material using manufacturer-recommended low voltages,⁵⁸ which leads to non-adiabatic conditions.⁵⁸

\textbf{PWM control}: Despite facing all the same challenges as Direct Joule Heating, PWM allows controlled power delivery through variation of duty cycle. When implemented in a closed-loop system, this setup utilizes energy more effectively and greatly reduces overheating and unnecessary power delivery.

\textbf{Capacitor Discharge} This activation method is also subject to similar losses as in direct joule heating;⁵⁹ however, by delivering extremely high initial⁵⁹

current (kiloamperes) in millisecond bursts, the SMA wire achieves a pseudo-adiabatic activation, allowing the phase transformation to occur before any significant heat loss occurs.

Induction:

This method is highly efficient, particularly for large-scale components, such as aerospace torque tubes. This is attributed to the direct volumetric heating and the reduced contact-based heat losses of resistive heating. The slow warm-up times of external heaters are reduced as heat generation occurs directly within the sample via eddy currents.

Response Time:

Direct Joule Heating: Typically, a slower activation method, its effectiveness is fundamentally dictated by the transient balance between the power supplied and the thermal resistance of the material and its environment. They often have a response in the range of seconds\cite{potapov2000time, guan2020experimental} when following standard recommended voltage levels.

PWM control: As PWM is a method to regulate Direct Joule Heating, it inherently has a slightly longer response time; however, using high duty cycles can bring the response time very close to that of an equivalent, unregulated Direct Joule Heating sample.

Capacitor Discharge

This method has an incredibly quick activation time due to the very high current being pulsed through the sample. The response time is so low that the sample reaches its transformation temperature (A_f) before any substantial heat can dissipate out from the SMA, which also adds to the efficiency of the control method.

\textbf{Induction}

Induction provides relatively rapid heating for large pieces regardless of geometry, bypassing the slow thermal conductivity limits of SMAs that hinder large-scale conductive heating. Volumetric generation of heat through induced eddy currents enables faster and more uniform heating in large/bulky samples. However, in smaller applications, the heating rate can become very slow, and the depth to which the magnetic field penetrates (skin depth) is very large compared to the cross-section of the sample. Additionally, the inductance of smaller wires is often negligible compared to that of the induction coil.

\subsection{Control and Precision:}

\textbf{Direct Joule Heating}:

Direct Joule Heating employs constant voltage or current to trigger phase transformation through internal Joule heating. It is characterized by low precision and slow response times (often several seconds) when using standard recommended voltages. Unregulated DC heating often results in irregular hysteresis shapes due to non-uniform temperature distribution and significant heat loss to the environment (non-adiabatic conditions).

\textbf{PWM control}

PWM regulates the time-averaged energy delivered to the SMA by means of duty cycle modulation. A microcontroller generates a train of high-frequency pulses (typically 1 kHz), varying the duty cycle $\text{\eqref{Ds}}$ to approximate a desired position.

It provides excellent software-selectable position tuning \cite{ma2017control, elahinia2012review}. Research on 100 μm wires shows that contraction begins at 50\% duty cycle, with major shifts at 70\% and completion at 80\%

\textbf{Capacitor Discharge}

Pulsed actuation using capacitors achieves ultra-fast activation by delivering high-amplitude current bursts in the millisecond range. Precision is often sacrificed for speed (as fast as 4.6 ms)\cite{qiu2000high}. Systems are highly sensitive to voltage thresholds, and if the charging voltage is too low, the transformation will not complete. There is a very limited scope for precision and control in Pulsed actuation systems.

\textbf{Induction}

Induction uses alternating current (AC) in a coil to generate an electromagnetic field, inducing internal eddy currents in the SMA. Performance is regulated ⁷⁴by the magnitude and frequency of the coil current, as well as coil geometry and the proximity of flux concentrators. It offers superior thermal control compared to resistive methods, with regulated chambers reaching a sensitivity of 0.01°C. This uniform, homogeneous heating produces stable ⁷⁵and prominent hysteresis loops, making it ideal for the precise thermomechanical characterization of larger components. It is the most effective method for large-scale components (e.g., aerospace torque tubes), where direct DC is current-limited and inefficient.

\subsection{Advantages}

\textbf{Direct Joule Heating}:

Direct Joule Heating is the most popular method of actuation for small-scale applications like ⁷⁶SMA wires and ribbons due to its very ⁷⁶simple implementation, requiring only a simple ⁷⁶controllable heating element and basic circuitry.

\textbf{PWM control}

PWM allows for accurate software-selectable control of length and displacement by adjusting the duty cycle to approximate desired output positions. It also consumes less energy than a constant DC voltage by delivering power-kicks that maintain the actuator's configuration without continuous full-power consumption.

\textbf{Capacitor Discharge}

Pulsed actuation using capacitors offers the shortest recorded response times, reaching full activation in as little as 4.6 to 6.5 milliseconds. By delivering energy so rapidly that transformation occurs before heat can dissipate to the environment, it achieves energy savings of up to 80\% compared to traditional quasi-static heating. Unlike regulated current methods, this can be performed with hardware capable of applying a single short voltage impulse, avoiding cost-intensive power electronics.

\textbf{Induction}

Induction uses electromagnetic fields to induce eddy currents directly within the SMA. This is a major advantage for medical implants (e.g., stents) and aerospace parts where physical electrical connections are impractical. It provides a highly uniform temperature distribution, which results in stable and predictable shape memory effect (SME) hysteresis loops. It is significantly more efficient than direct DC for large-scale components (e.g., torque tubes), requiring much lower current to achieve the same heating rate.\cite{dutta2019review, el2020shape}

\section{Trade-offs}

The different methods of electrical activation for shape memory alloys (SMAs) each possess specific drawbacks, ranging from energy inefficiencies to complexities in mechanical control.

\subsection{Direct DC (Standard Resistive/Joule Heating)}

\begin{itemize}

\item \textbf{Response Speed:} This method typically results in a slow actuation response (on the order of seconds) because of the long time required to heat the material using manufacturer-recommended low voltages.

\item \textbf{Energy Efficiency:} It often occurs under non-adiabatic conditions, meaning a large portion of the heating energy is lost to the ⁷⁹environment through convection and conduction.

\item \textbf{Thermal Control:} Unregulated direct heating can lead to thermal inhomogeneity, creating uneven surface temperature distributions and irregular hysteresis shapes.

\item \textbf{Safety Risks:} Applying standard industrial voltage levels continuously can result in very high currents that may overheat and damage or destroy the SMA actuators.

\end{itemize}

\subsection{Pulse Width Modulation (PWM)}

\begin{itemize}

\item \textbf{Implementation Complexity:} Unlike basic DC activation, PWM requires more complex electronic hardware, including microcontrollers (like the 8-bit PIC) and MOSFET drivers to handle high-frequency switching.

\item \textbf{Speed Control Limits:} It is extremely difficult to control the displacement speed of the wire when using duty cycles between 75\% and 85\%, as the material enters its austenitic phase almost instantaneously (approximately 0.1 seconds) in this range.

\item \textbf{Lag and Hysteresis:} PWM systems suffer from a thermal mass lag, which is a delay between the input current increase and the resulting force output. Additionally, slow cooling rates can cause pronounced hysteresis (10-20\%) in temperature and displacement characteristics.

\item \textbf{Cooling Bottlenecks:} If used at high frequencies without a cooling source, the actuator can accumulate heat, leading to high position errors in the cooling stage.

\end{itemize}

\subsection{Capacitor Discharge (High-Voltage Pulsed Heating)}

\begin{itemize}

\item \textbf{"Ballistic" Behavior:} The nearly instantaneous force generation can accelerate the attached mass so intensely that the SMA wire becomes slack, causing a "ballistic" stroke where displacement is driven by inertia rather than controlled transformation.

\item \textbf{Circuit Sensitivity:} Systems are highly sensitive to voltage thresholds; for a given capacitance, if the charging voltage is set below the⁸¹ "lower bound" (V_L), the energy will be insufficient to complete the phase transformation.

\item \textbf{Mechanical Stability:} High-speed activation frequently causes the mass-spring-damper system to oscillate before reaching a stable state, which may be undesirable for certain proportional tasks.

\item \textbf{Repeatability Issues:} Variations in contact and transition resistance caused by manual clamping of the SMA wires can lead to inconsistent heating rates between different activation cycles.

\end{itemize}

\subsection{Induction Heating (Contactless AC)}

\begin{itemize}

\item \textbf{Skin Effect:} At high frequencies, 86\% of the heating power is concentrated on the surface of the workpiece, leaving the core to be heated primarily through slower internal conduction.

\item \textbf{Size Limitations:} Induction heating is inefficient for thin wires; the heating rate becomes extremely slow because the skin depth is large relative to the wire diameter and the wire's inductance is negligible.

Equipment Requirements: This method requires bulky and specialized equipment, such as large electromagnets and high-frequency generators, which can be difficult to integrate into compact systems.

Geometry Sensitivity: To avoid "cold corners" or non-uniform transformation, the induction coil must be precisely matched to the shape of the SMA component.⁸²

Results and Discussion

To evaluate the different SMA activation methods, numerical simulations were carried out using the lumped thermal model described earlier⁸³

⁸⁴[\cite{huang2012optimisation}](#). The temperature response of the SMA was studied under different electrical inputs corresponding to direct Joule heating,⁸⁵ PWM control, capacitor discharge, and induction heating.

Temperature-Time Response

⁸⁶Temperature of the SMA changes over time for the different activation⁸⁷ methods. Direct Joule heating shows a gradual increase in temperature, with response times typically on the order of seconds. This is mainly due to⁸⁸ continuous heat loss to the surroundings during heating.

PWM-controlled heating follows a similar overall trend, but the rate of temperature rise depends on the duty cycle. Higher duty cycles result in faster heating, while lower duty cycles slow the response.

In contrast, capacitor discharge results in a very rapid temperature increase, reaching the transformation temperature within milliseconds.⁸⁹ This is due to the very high current delivered in a short time, which leads to near pseudo-⁹⁰adiabatic heating where heat loss during the activation period is minimal.⁹⁰

Induction heating shows relatively fast heating for larger samples due to volumetric heat generation through eddy currents. However, for smaller samples, the heating rate becomes slower due to reduced coupling efficiency.

`\subsection{Steady-State Behaviour}`

The steady-state temperature for each method follows the relation given in `\eqref{sstate}`. For PWM control, a roughly linear relationship between duty cycle and steady-state temperature is observed,⁹¹ showing that $T_{ss} \propto D$.

Capacitor discharge does not have a steady-state behaviour⁹² in the usual sense, since it operates through short pulses. Instead, the peak temperature reached depends on the initial capacitor voltage.

Induction heating shows a nonlinear⁴ relationship with input parameters, with temperature approximately scaling with the square of the magnetic field strength.

`\subsection{Discussion of Trade-offs}`

From the results, it is clear that there are trade-offs between response time, efficiency, and control. PWM provides the most stable and controllable heating,⁹³ while capacitor discharge gives the fastest response.⁹³ Induction heating works well for larger systems but is less effective for smaller geometries.

Overall, the results follow the expected trends from the theoretical analysis,⁹⁴ and show that the choice of activation method depends on the requirements of the specific application.

\section{Conclusion}

This paper compared different electrical activation methods for shape memory alloys, including direct Joule heating, PWM, capacitor discharge, and induction-based approaches. Each method affects the behaviour of the SMA differently,⁹⁵ especially in terms of response time, efficiency, and how easy it is to control.⁹⁶

From the comparison, it can be seen that direct Joule heating is simple but not very efficient and difficult to control precisely.^{97,99} PWM improves this by allowing better control over the heating process through duty cycle variation,⁹⁸ making it more suitable for applications where precision is important.¹⁰⁰ Capacitor discharge provides the fastest response due to the very high current delivered in a short time, but this also makes it harder to control and less stable in some cases.⁹⁸ Induction heating works well for larger systems,¹⁰¹ as it provides efficient and uniform heating,¹⁰¹ but it is not as effective for smaller SMA elements.¹⁰¹

Overall, no single method performs best in all cases. There is always a trade-off between speed, efficiency, and control. Because of this, the choice of activation

method depends on the specific application and what parameter is most important for that system.

```
\bibliographystyle{IEEEtran}
```

```
\bibliography{references}
```

```
\end{document}
```

1.	<i>Shape memory alloys (SMAs) are materials that possess the ability to return to their original shape when exposed to stimuli such as temperature changes.</i>	Paragraph can be improved	Clarity
2.	<i>This</i>	Intricate text	Clarity
3.	<i>Properties such as Shape Memory Effect (SME) and superelasticity (SE) are also a result of this transition.</i>	Paragraph can be improved	Clarity
4.	<i>non-linear; nonlinear</i>	Text inconsistencies	Correctness
5.	<i>In practice, the behavior of SMAs is non-linear and hysteretic, and their control is complex.</i>	Paragraph can be improved	Clarity
6.	<i>is limited</i>	Passive voice misuse	Clarity
7.	<i>This paper compares different electrical activation methods for SMAs, including direct Joule heating, PWM, capacitor-discharge-based pulsed heating, and electromagnetic methods such as induction.</i>	Paragraph can be improved	Clarity
8.	<i>Each method is evaluated under similar conditions so that the comparison is fair, and the results are considered in terms of efficiency, response time, and how easy the system is to control.</i>	Paragraph can be improved	Clarity
9.	<i>The results show that PWM has the highest control, capacitor discharge has the fastest actuation, and induction offers the highest efficiency in regard to large-scale systems.</i>	Paragraph can be improved	Clarity
10.	behaviour → behavior	Mixed dialects of English	Correctness

11.	<i>This behaviour occurs due to a solid-to-solid phase transformation between two different crystal structures, known as austenite and martensite \cite{lagoudas2008shape, brinson1993one}.</i>	Paragraph can be improved	Clarity
12.	behaviour → behavior	Mixed dialects of English	Correctness
13.	<i>This</i>	Intricate text	Clarity
14.	<i>This was followed by Greninger in 1938, who observed the memory effect, which was later described more clearly by Vernon in 1941 as shape memory.</i>	Passive voice misuse	Clarity
15.	<i>This was followed by Greninger in 1938, who observed the memory effect, which was later described more clearly by Vernon in 1941 as shape memory.</i>	Paragraph can be improved	Clarity
16.	behaviour → behavior	Mixed dialects of English	Correctness
17.	<i>The response is non-linear and hysteretic \cite{shaw2008thermomechanical, liu1998hysteresis}, which makes it difficult to control accurately.</i>	Paragraph can be improved	Clarity
18.	<i>In addition to this, the actuation speed is limited by the rate of heating and cooling, meaning that SMAs often have a relatively slow response compared to other actuator systems \cite{huang2012optimisation}.</i>	Paragraph can be improved	Clarity
19.	<i>These limitations reduce their effectiveness in applications where fast or precise control is required.</i>	Paragraph can be improved	Clarity
20.	<i>are used</i>	Passive voice misuse	Clarity

21.	<i>is delivered</i>	Passive voice misuse	Clarity
22.	behaviour → behavior	Mixed dialects of English	Correctness
23.	<i>Each of these methods affects the behaviour of the SMA in different ways, particularly in terms of efficiency, response time, and how easy it is to control.</i>	Paragraph can be improved	Clarity
24.	<i>is defined</i>	Passive voice misuse	Clarity
25.	<i>is applied</i>	Passive voice misuse	Clarity
26.	<i>is removed</i>	Passive voice misuse	Clarity
27.	<i>The transformation between these two phases is governed by four characteristic temperatures:</i>	Passive voice misuse	Clarity
28.	<i>is completed</i>	Passive voice misuse	Clarity
29.	<i>are not fixed</i>	Passive voice misuse	Clarity
30.	<i>is released</i>	Passive voice misuse	Clarity
31.	<i>In high-speed actuation, such as capacitor discharge, delivering energy rapidly enough to overcome this latent heat threshold is essential for achieving adiabatic activation, which minimizes total energy loss.</i>	Paragraph can be improved	Clarity
32.	<i>The current and voltage relation is given by Ohm's law $\text{\eqref{ohmslaw}}$, and the resistance for a uniform material with resistivity ρ_R and constant current flow is given by $\text{\eqref{eq:rho}}$.</i>	Passive voice misuse	Clarity
33.	<i>be predicted</i>	Passive voice misuse	Clarity

34.	<i>I</i>	Inappropriate colloquialisms	Delivery
35.	<i>If instead the applied voltage is held constant, the current becomes inversely proportional to the wire length, $I = \frac{V}{R} \propto \frac{1}{L}$, leading to different thermal behavior.</i>	Paragraph can be improved	Clarity
36.	<i>Pulse Width Modulation (PWM) is a control technique that utilizes a periodic train of electrical pulses where the base frequency remains fixed, but the width of each pulse is varied.</i>	Paragraph can be improved	Clarity
37.	<i>is used</i>	Passive voice misuse	Clarity
38.	<i>In the context of shape memory alloys (SMAs), it is used to precisely regulate the amount of energy transferred to the material to control its temperature, length, and displacement.</i>	Paragraph can be improved	Clarity
39.	<i>the average power delivered to the SMA can be expressed</i>	Passive voice misuse	Clarity
40.	<i>the heating rate can be directly regulated</i>	Passive voice misuse	Clarity
41.	<i>Capacitor; capacitor</i>	Text inconsistencies	Correctness
42.	<i>This</i>	Intricate text	Clarity
43.	<i>I</i>	Inappropriate colloquialisms	Delivery
44.	<i>This method is uniform in larger-scale applications and has the ability to be regulated precisely.</i>	Paragraph can be improved	Clarity
45.	<i>This</i>	Intricate text	Clarity

46.	<i>Induction systems often feature feedback controllers with high temperature sensitivity, and because the heating is uniform and homogenous, induction-based SMAs produce stable and prominent hysteresis loops.</i>	Paragraph can be improved	Clarity
47.	<i>For instance, a 13.56 MHz field can trigger one section while a 296 kHz field triggers another.</i>	Paragraph can be improved	Clarity
48.	<i>is raised</i>	Passive voice misuse	Clarity
49.	<i>is generated</i>	Passive voice misuse	Clarity
50.	<i>Because heat is generated inside the polymer itself, magnetic activation allows for a rapid temperature increase with a reduced thermal gradient compared to external heating.</i>	Paragraph can be improved	Clarity
51.	<i>is achieved</i>	Passive voice misuse	Clarity
52.	<i>In order to compare the different methods previously outlined in a fair manner, certain assumptions are made, which will be specified.</i>	Paragraph can be improved	Clarity
53.	<i>are made</i>	Passive voice misuse	Clarity
54.	<i>is assumed</i>	Passive voice misuse	Clarity
55.	<i>is based</i>	Passive voice misuse	Clarity
56.	<i>is given</i>	Passive voice misuse	Clarity
57.	<i>Although SMA activation methods are primarily thermal in nature, they vary in how the heat is regulated and delivered.</i>	Paragraph can be improved	Clarity

58.	<i>This low efficiency is a direct result of the long durations required to heat the material using manufacturer-recommended low voltages, which leads to non-adiabatic conditions.</i>	Paragraph can be improved	Clarity
59.	<i>\textbf{Capacitor Discharge} This activation method is also subject to similar losses as in direct joule heating; however, by delivering extremely high initial current (kiloamperes) in millisecond bursts, the SMA wire achieves a pseudo-adiabatic activation, allowing the phase transformation to occu...</i>	Paragraph can be improved	Clarity
60.	<i>This</i>	Intricate text	Clarity
61.	<i>is attributed</i>	Passive voice misuse	Clarity
62.	<i>This is attributed to the direct volumetric heating and the reduced contact-based heat losses of resistive heating.</i>	Paragraph can be improved	Clarity
63.	<i>are reduced</i>	Passive voice misuse	Clarity
64.	<i>The slow warm-up times of external heaters are reduced as heat generation occurs directly within the sample via eddy currents.</i>	Paragraph can be improved	Clarity
65.	<i>is fundamentally dictated</i>	Passive voice misuse	Clarity
66.	seconds\cite → seconds \cite	Misspelled words	Correctness
67.	<i>They often have a response in the range of seconds\cite{potapov2000time, guan2020experimental} when following standard recommended voltage levels.</i>	Paragraph can be improved	Clarity

68.	<i>being pulsed</i>	Passive voice misuse	Clarity
69.	<i>The response time is so low that the sample reaches its transformation temperature (T_A) before any substantial heat can dissipate out from the SMA, which also adds to the efficiency of the control method.</i>	Paragraph can be improved	Clarity
70.	<i>Volumetric generation of heat through induced eddy currents enables faster and more uniform heating in large/bulky samples.</i>	Paragraph can be improved	Clarity
71.	<i>However, in smaller applications, the heating rate can become very slow, and the depth to which the magnetic field penetrates (skin depth) is very large compared to the cross-section of the sample.</i>	Paragraph can be improved	Clarity
72.	<i>A microcontroller generates a train of high-frequency pulses (typically 1 kHz), varying the duty cycle D_s to approximate a desired position.</i>	Paragraph can be improved	Clarity
73.	<i>→ .</i>	Incorrect punctuation	Correctness
74.	<i>is regulated</i>	Passive voice misuse	Clarity
75.	<i>This uniform, homogeneous heating produces stable and prominent hysteresis loops, making it ideal for the precise thermomechanical characterization of larger components.</i>	Paragraph can be improved	Clarity
76.	<i>Direct Joule Heating is the most popular method of actuation for small-scale applications like SMA wires and ribbons due to its very simple implementation, requiring only a simple controllable heating element and basic circuitry.</i>	Paragraph can be improved	Clarity

77.	<i>PWM allows for accurate software-selectable control of length and displacement by adjusting the duty cycle to approximate desired output positions.</i>	Paragraph can be improved	Clarity
78.	<i>This</i>	Intricate text	Clarity
79.	<i>is lost</i>	Passive voice misuse	Clarity
80.	<i>is driven</i>	Passive voice misuse	Clarity
81.	<i>is set</i>	Passive voice misuse	Clarity
82.	<i>the induction coil must be precisely matched</i>	Passive voice misuse	Clarity
83.	<i>were carried out</i>	Passive voice misuse	Clarity
84.	<i>The temperature response of the SMA was studied</i>	Passive voice misuse	Clarity
85.	corresponding to → :	Paragraph can be improved	Clarity
86.	Temperature → The temperature	Determiner use (a/an/the/this, etc.)	Correctness
87.	<i>Temperature of the SMA changes over time for the different activation methods.</i>	Paragraph can be improved	Clarity
88.	<i>This</i>	Intricate text	Clarity
89.	<i>This</i>	Intricate text	Clarity
90.	<i>This is due to the very high current delivered in a short time, which leads to near pseudo-adiabatic heating where heat loss during the activation period is minimal.</i>	Paragraph can be improved	Clarity

91.	<i>is observed</i>	Passive voice misuse	Clarity
92.	behaviour → behavior	Mixed dialects of English	Correctness
93.	<i>PWM provides the most stable and controllable heating, while capacitor discharge gives the fastest response.</i>	Paragraph can be improved	Clarity
94.	analysis,	Comma misuse within clauses	Correctness
95.	behaviour → behavior	Mixed dialects of English	Correctness
96.	<i>Each method affects the behaviour of the SMA differently, especially in terms of response time, efficiency, and how easy it is to control.</i>	Paragraph can be improved	Clarity
97.	<i>it can be seen</i>	Passive voice misuse	Clarity
98.	<i>From the comparison, it can be seen that direct Joule heating is simple but not very efficient and difficult to control precisely.</i>	Paragraph can be improved	Clarity
99.		Tone suggestions	Delivery
100.	<i>PWM improves this by allowing better control over the heating process through duty cycle variation, making it more suitable for applications where precision is important.</i>	Paragraph can be improved	Clarity
101.	<i>Induction heating works well for larger systems, as it provides efficient and uniform heating, but it is not as effective for smaller SMA elements.</i>	Paragraph can be improved	Clarity

102.	<i>possess the ability to return to their original shape</i>	Describe the types of materials used in biomedical engineering, and their applications - My Exam Solution https://www.myexamsolution.com/2023/06/describe-the-types-of-materials-used-in-biomedical-engineering-and-their-applications.html	Originality
103.	<i>Properties such as Shape Memory Effect (SME) and superelasticity (SE)</i>	ABM Proceedings - INFLUÊNCIA DA VARIAÇÃO DE PARÂMETROS NA PRODUÇÃO IN-SITU DA LIGA NITI POR FUSÃO SELETIVA A LASER https://abmproceedings.com.br/pibr/article/influncia-da-variao-de-parmetros-na-produo-in-situ-da-liga-niti-por-fuso-seletiva-a-laser-1	Originality
104.	<i>the shape memory effect (SME) and superelasticity (SE)</i>	The Potential of Shape Memory Alloys in Riveting Applications	Originality
105.	<i>limited by the rate of heating and cooling, meaning</i>	Meet this robotic fish with fins which may help in underwater exploration https://interestingengineering.com/innovation/robotic-fish-with-fins-underwater-exploration	Originality
106.	<i>deformation that remains after the load is removed.</i>	What is Deformation? — Kreo Glossary https://www.kreo.net/glossary/deformation	Originality
107.	<i>occur at the same temperature during heating and cooling,</i>	Spinodal decomposition influence of austenite on martensitic transition in a Mn-13 at.%Cu alloy	Originality

-
- | | | |
|------|---|-------------|
| 108. | <i>dynamics}. \begin{itemize} \item
 \textbf{Definition:} Hysteresis is the
 temperature difference between the
 heating and cooling paths, often
 characterized by the difference
 between the temperatures at which
 50\% of the material has transformed
 during each cycle. \item
 \textbf{Performance:} A narro...</i> | Originality |
|------|---|-------------|
-
- | | | |
|------|---|-------------|
| 109. | <i>and the resulting force increase. In
 high-speed actuation, such as
 capacitor discharge, delivering energy
 rapidly enough to overcome this
 latent heat threshold is essential for
 achieving adiabatic activation, which
 minimizes total energy
 loss.\cite{miyazaki1989sma}
 \end{itemize} \section{Activati...</i> | Originality |
|------|---|-------------|
-
- | | | |
|------|--|-------------|
| 110. | <i>n2005heat, zhou2010sma}
 \subsection{PWM Control} Pulse
 Width Modulation (PWM) is a control
 technique that utilizes a periodic
 train of electrical pulses where the
 base frequency remains fixed, but the
 width of each pulse is varied.</i> | Originality |
|------|--|-------------|
-
- | | | | |
|------|---|---|-------------|
| 111. | <i>In the context of shape memory alloys
 (SMAs</i> | Innovative Spring Designs
Revolutionizing the Manufacturing
Industry: Shape Memory Springs -
Western Spring
https://www.westernspring.com/western-spring-resources/innovative-spring-designs-revolutionizing-the-manufacturing-industry-shape-memory-springs/ | Originality |
|------|---|---|-------------|
-
- | | | |
|------|---|-------------|
| 112. | <i>e applications and has the ability to
 be regulated precisely. Induction
 utilizes Faraday's Law to induce eddy
 currents directly within the
 conductive SMA. This generates heat
 from within the material, eliminat</i> | Originality |
|------|---|-------------|
-

113.	<i>hape recovery by generating heat directly within the material. Magnetic activation relies on inductive heating, where the material's internal temperature is raised through three primary electromagnetic mechanisms: Eddy Current Losses: Dominant in larger, electrically conductive components. Hystere...</i>		Originality
114.	<i>the heat transfer coefficient, h is the temperature</i>	High yields of hydrogen production from methanol steam reforming with a cross-U type reactor	Originality
115.	<i>n dissipate out from the SMA, which also adds to the efficiency of the control method. \textbf{Induction} Induction provides relatively rapid heating for large pieces regardless of geometry, bypassing the slow thermal conductivity limits of SMAs that hinder large-scale conductive heating. Volu...</i>		Originality
116.	<i>gible compared to that of the induction coil. \subsection{Control and Precision:} \textbf{Direct Joule Heating}: Direct Joule Heating employs constant voltage or current to trigger phase transformation through internal Joule heating. It is characte</i>		Originality
117.	<i>by the magnitude and frequency of the coil current, as well as coil geometry and the proximity of flux concentrators. It offers superior thermal control compared to resistive methods, with regulated chambers reaching a sensitivity of 0.01°C. This uniform, homogeneous heating produces stable a</i>		Originality

118.	<i>achieves energy savings of up to 80\% compared to traditional</i>	Fusion Air Lighting Partners HVAC and Lighting Integration ARCAT https://www.arcat.com/company/fusion-air-lighting-partners-53945?pids=227104	Originality
119.	<i>nse (on the order of seconds) because of the long time required to heat the material using manufacturer-recommended low voltages. \item \textbf{Energy Efficiency:} It often occurs under non-adiabatic conditions, meaning a large portion of the heating energy is lost to the environment through conve...</i>		Originality
120.	<i>s that may overheat and damage or destroy the SMA actuators. \end{itemize} \subsection{Pulse Width Modulation (PWM)} \begin{itemize} \item \textbf{Implementation Complexity:} Unlike basic DC activation, PWM requires mo</i>		Originality
121.	<i>o high position errors in the cooling stage. \end{itemize} \subsection{Capacitor Discharge (High-Voltage Pulsed Heating)} \begin{itemize} \item \textbf{"Ballistic" Behavior:} The nearly instantaneous force generation can accelerate the attached mass so intensely th</i>		Originality